

Basic distinctions in the AI power consumption debate

Posted by u/Human_certified • 0 points • 12 comments

Whatever side you're arguing, these distinctions matter.

I included the most commonly accepted figures as of May 2025. [Only boring sources used (IEA, Gartner)](<https://www.iea.org/reports/energy-and-ai/energy-demand-from-ai>). There is quite some uncertainty in all of them, but if someone else's figures are **wildly** off compared to these, that should be a red flag or (more likely) they're just not referring to the same thing.

****US $\neq$ the world**** \- Seems incredibly obvious, but I see US and global figures being confused all the time. All figures in this post are **global.**

****Electricity $\neq$ energy -**** Electricity accounts for around ****20%**** of all global end-use energy. Its proportion is rising, but most end-use energy is still fuel or heat. Even reliable sources seem to screw up global electricity vs. energy all the time, and thus end up being randomly off by a factor 5x.

****Data center power $\neq$ total electricity**** \- Data centers currently account for around ****1.5%**** of global electricity use. They represent the bulk of our non-desktop computing power use, hosting all of our cloud, web, social media, communications, gaming, business, and streaming services. (Data center power consumption has been rising steadily and steeply for decades, representing a shift to online and cloud computing and increased use of IT by people worldwide.)

****AI/ML power $\neq$ data center power -**** Artificial intelligence (AI) or machine learning (ML) today accounts for around ****20%**** of global data center power. AI/ML includes social media algorithms, product recommendations, speech recognition, image and video captioning, spam filtering, and threat detection heuristics, which are power-hungry due to the sheer scale of the data they have to process.

****GenAI power $\neq$ AI/ML power**** \- Generative AI (ChatGPT, DeepSeek, Midjourney, Veo3) is currently around ****25-50%**** of all AI/ML power, but is rising steeply. It depends on what is considered "generative". By sheer volume, this can be assumed to be predominantly LLMs (chatbots), not image or video generation (with the latter mostly being limited or paywalled).

****Training power $\neq$ inference power**** \- Training (one-off) is estimated to account for

****20%**** of GenAI power use, with inference (actually using the AI to generate something) the remaining ****80%****.

Naively multiplying the above, we find that ****generative AI today accounts for 0.015%-0.03% of global end-use energy, but it's rising steeply****. This calculation puts GenAI energy use well *above* most other estimates (0.005%-0.013%). Presumably someone's definitions got messed up somewhere, or uncertainties added up?

The IEA "best estimates" for data center growth are around 30% per year for "accelerated servers" (i.e. those using GPUs, not just for AI). Using that figure^{**}, the IEA arrives at a share of around 0.028% of global energy consumption by GenAI by 2030,^{**} close to the high end of the above naive calculation for 2025.

There have, however, been recent articles that lead to figures like "4% of global electricity". Depending on definitions, that's 30-150 times (!) more than the IEA now projects.

(This is where you might want to apply a sanity test and consider your source, and your source's sources.)

Finally, ****weird comparisons don't help anyone and just make you sound dodgy**** \- "If US AI power consumption quadruples by the end of 2028, then global data center power consumption in a single month could be as much as all of Nepal uses in seven years, or enough to power four Eiffel Towers for half a century." (Also, people are as likely as not to say: "Oh, is that all?")

Comments

u/Exciting-Interest820 • 1 points • 2025-06-03 20:03 UTC

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u/Tyler_Zoro • 2 points • 2025-06-03 18:36 UTC

> AI/ML power $\neq$ data center power - Artificial intelligence (AI) or machine learning (ML) today accounts for around 20% of global data center power.

I think that even that number is inflated by focusing primarily on cloud compute hosting providers who are used for AI modeling disproportionately. Let's first remember that the largest amount of compute in the world is almost certainly hidden behind black budgets. We know that the NSA maintains extremely large datacenters for performing surveillance analysis, but we have little idea what processing is happening there.

Next, keep in mind that thousands of companies have self-hosted or non-cloud hosted server

infrastructure. Most of that infrastructure has nothing to do with AI, and your 20% figure would almost certainly be far too high in those cases.

u/jay-ff • 1 points • 2025-06-03 20:21 UTC

I think people that make estimates have ways to take these things into account, maybe not every nsa data center but I doubt that they make a dent without showing up somewhere—especially because the largest “conventional” supercomputers are known, even the ones that probably have dual use.

u/Tyler_Zoro • 1 points • 2025-06-03 20:30 UTC

> I think people that make estimates have ways to take these things into account

I would certainly not make such assumptions. If you can demonstrate that such issues are taken into account, I'll listen, but I've worked in that industry too long to believe the trade press on datacenter allocation.

> the largest “conventional” supercomputers are known

I have no idea why you're talking about supercomputers. They haven't really been relevant for over a decade, arguably multiple decades. The present compute economy is based around large collections of small systems. AI has changed that a bit, because right now AI models need staggeringly large amounts of VRAM in each system, but even given that, there's no supercomputer running the show, but a collection of systems that all have top-end GPUs.

u/TheJzuken • 2 points • 2025-06-03 17:47 UTC

Same goes for water use. "Oh no, generating a single images takes a bottle of water!" - mate, perchance you also shower, flush, wash your hands, dishes and cloths, but I wouldn't put my faith in that with some antis.

u/DKMK_100 • 2 points • 2025-06-03 22:49 UTC

Wait till they learn how many gallons of water it takes to make one hamburger. That's a lot of bottles...

u/The_rule_of_Thetra • 3 points • 2025-06-03 20:29 UTC

Besides, the faulty plumbing system in the US consumes an amount of water larger than AI generations by the hundreds of thousands more each day.

And one can be quite easily fixed and would also improve the quality of the water Americans use each day, but alas...

u/Person012345 • 4 points • 2025-06-03 15:35 UTC

All told, how much of a percent of global CO2 emissions does AI generate? And how does that percent change if countries stick to clean energy plans?

And more to the point, why does this matter to someone in iceland generating pictures on his local computer from practically entirely renewable sources? How is he a worse person than some crybaby american anti who drives to the grocery store then drives home to their air conditioned house?

u/The_rule_of_Thetra • 13 points • 2025-06-03 15:28 UTC

Let's also add another important distinction, shall we?

****Local Generation\\Training != Big Company generation\\Training****

Like, hate me all you want, mates... but if you compare someone like me, that makes everything local (both generations and LORA training), who also uses solar panels to power up everything in his home, to what companies like Open-AI do, then you are not an idiot... You are spreading fake news simply to take the moral high ground. Heck, my PC consumed more energy when I was playing Claire Obscure: Expedition 33 with all in Ultra than what I was consuming when I was generating myself some tiddies.

As OP said: weird comparisons don't help anyone and just make you sound dodgy... but bring up fake news and get caught red-handed doing so? Then you aren't simply dodgy, you will be mistrusted and even scorned. Do things the right way, or don't do it all, unless you wish to light up torches and burn your entire house down with your blind anger.

u/Superseaslug • 2 points • 2025-06-05 02:09 UTC

"generating myself some tiddies"

![gif](giphy|6xFN8KIGD56Dnyutel)

u/Electric-Molasses • 2 points • 2025-06-03 18:26 UTC

I didn't even realize people would go after independent enthusiasts, that just sounds wack to me. You're heavily using a personal computer LOL

u/AutoModerator • 1 points • 2025-06-03 15:17 UTC

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